

The SAFe way to the Agile Organization

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ABSTRACT

How do you make the agile organization? This paper gives an answer in form of a case story from the financial software company SimCorp. In late 2015 SimCorp decided to go agile on a big scale. To do so they applied the scalable agile framework SAFe. In this framework Scrum teams are working within “Agile Release Trains” providing value streams that together realize a strategic mission. At SimCorp there were more than 500 employees in 55 teams working in eight agile release trains. SimCorp has a solid track record of releasing a new version of their standard software product – SimCorp Dimension – every 6 months. In February 2017 a new version of the product was successfully released according to plan and now with an agile organization behind it. The paper tells the story from beginning to end focusing on the three things that made it possible to become an agile organization within a year. Furthermore the paper discusses what the main challenges were.

CCS CONCEPTS

Software and its engineering → **Software creation and management** → **Software development process management** → **Software development methods** → **Agile software development**

KEYWORDS

Agile software development; large-scale software development; scaled agile framework

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1 INTRODUCTION

Agile development of software and agile projects has been around for the last 20 years. Around the turn of the century this new way of working had many names such as “internet speed development” but with the agile manifesto in 2001 the common denominator for the new way of developing software became agile.

Over the years methods such as XP and Scrum became popular. In 2015 a Danish study of close to 600 project managers (Cordua et al. 2015) showed that 40% of all Danish and international projects in all domains were using agile methods. And of these projects using agile methods more than half were using Scrum.

An interesting question now that agile has become so widespread among projects is how to scale agile up to use it throughout an organization?

Version One (2016: p. 13) report that the most commonly used way to scale agile is to use Scrum and Scrum-of Scrums (72%). The second most commonly used way is SAFe – Scale Agile Framework originally developed by Leffingwell (2007) and now in version 4.0 (Leffingwell 2017). The current version of SAFe is thoroughly described in a White Paper (scaled Agile 2017); here is a short excerpt from that:

“SAFe is based fundamentally on Agile teams. Each team is responsible for defining, building, and testing stories ...SAFe teams are organized into a virtual program structure called the “Agile Release Train” (ART). Each ART is a long-lived, self-organizing team of Agile teams (typically 5 to 12) ... The Portfolio level organizes and funds a set of value streams. The value streams realize a set of solutions, which help the enterprise achieve its strategic mission...” (Scaled Agile 2017, p. 5)

In this paper we describe a case from the financial company SimCorp that decided to go agile on a big scale in early 2016 and launched their first agile version of the key product Dimension in February 2017.

2 SIMCORP

SimCorp is an internationally oriented company with more than 1,250 top qualified employees of which 90% hold an academic degree in economics, finance, or IT. SimCorp work closely with 170+ clients around the world, helping them to manage over USD 19 trillion in assets under management. Almost half (46%)

of the world's top 100 asset managers have selected the key product SimCorp Dimension, a front-to-back investment management solution.

Each year SimCorp use more than 20% of their revenue for R&D, ensuring that their SimCorp Dimension product is capable of supporting key requirements from customers.

3 RESEARCH METHOD

The method applied to write this paper can best be characterized as an action case (Vidken & Braa, 1997) or a participatory action research undertaking (Whyte 1991). One author was employed in SimCorp throughout the change to agile and was responsible for planning and carrying out the organizational change or the transformation as it was called. Thus the action part of the case was primarily carried out by this author. The other author has followed SimCorp as a researcher on and off since 1997. For this specific case meetings and interviews were conducted by this other author in January (full day), February (half day) March (two interviews lasting 1-2 hours), April (one meeting and one interview), and May 2017 (one interview). The first meeting in January was used to elicit a semi-structured interview guide. Here is an example of the content of this guide:

4. The Implementation

People I have talked with have mentioned your commitment and motivating as crucial. Also that you allowed for uncertainty by saying "I don't have all the answers". What are your thoughts on that?

Looking back at the whole implementation what were the 3 best things that others can learn from?

And what are 3 things you would do different today?

If we look at the details of SAFe. What has worked as-is and what have you had to change or adapt?

These questions were used for a 1.5 hour interview with the Chief Technical Officer of SimCorp on 29 May 2017. Similar semi-structured guides were used in the following interviews.

Interviews were recorded and summarized into detailed interview notes shortly afterwards. The following analysis built a "logical chain of evidence" and used analytic induction (Miles et al. 2014, p. 290ff). Finally the findings were synthesized into this paper.

4 CASE STORY - THE SAFe ORGANIZATIONAL CHANGE

In late 2015 SimCorp decided to change from plan-driven waterfall-like software development to agile development using SAFe version 4.0. About 500 software developers, testers and project managers in four countries (Mainly Denmark and Ukraine; small groups in UK and Germany) were to change the way they worked.

This major organizational change was led by the Chief Technical Officer (CTO) whom came to SimCorp seven years earlier from Germany.

In late 2015 the issues that came out in retrospectives at SimCorp were:

- A. No way to improve systematically
- B. Too little visibility
- C. Too early commitment to a design that didn't work
- D. Late delivery of new functionality
- E. Problems discovered too late
- F. Under-estimated dependencies
- G. Growth in complexity of SimCorp Dimensions software
- H. Hard to manage distributed teams
- I. Phase-gate software delivery life-cycle did not help reduce risks
- J. Poor morale among software developers in some places

Based on these issues it was decided to go "all in" on agile meaning that every software developer within SimCorp should change the way they worked; SimCorp were to become an agile organization.

After looking at different possibilities it was decided to use SAFe. Software development at SimCorp was divided into eight "Agile Release Trains". The first train started in April 2016 for a product stream called "Middle Office". The second train followed a few months after and the 8th and last train was initiated in January 2017.

5 THREE MAJOR CHALLENGES

Looking back at the whole process after the release of the first version of SimCorp Dimension the three major challenges had been:

1. The whole re-organization to an agile organization. SimCorp used SAFe as a blueprint for reorganizing. This meant that old jobs as line managers and project managers became obsolete. The former managers had to apply for SAFe jobs such as "train conductor", "portfolio manager" and "scrum master". There was no direct mapping from the old job roles to the new job roles. Thus some of the powerful line managers that before had responsibility for both software delivery and people had to give up one or the other. In March 2017 some individuals were still challenged with this change of role.

2. Implementing the agile mindset. Conveying the message that it is OK to let go and say goodbye to the "old ways". Agile is all about transparency and about continuously improving based on experiments or on failures. Being open about this requires courage and a change of mindset

3. The interfaces to other parts of the SimCorp organization. It was 'only' the 500 people working in the software development part of the organization that changed to agile following SAFe. But it's challenging to go full scale agile if supporting functions, e.g. Sales and Marketing, HR or IT, do not work in an agile way. Thus after the first release of SimCorp Dimension build in an agile way the second part of the

organizational transformation will be to include and change other parts of SimCorp such as Sales and Marketing. This is now (June 2017) happening also because customers of SimCorp have started asking for agile.

6 LEARNING FROM THE CASE

In a semi-structured interview we identified three pieces of advice and two things that SimCorp would do differently today. The three good pieces of advice were:

1. Management support and sponsorship are crucial. In this case it was very important that the CTO of SimCorp took the lead and stood on the soap box motivating for the change to an agile organization

2. Break out of certainty and change to the “start-doing” mindset. The CTO of SimCorp was known for always being analytic, having answers and being attentive to details. However, in implementing agile he allowed for uncertainty; “I don’t have all the answers”, he said.

3. You need the right team to implement agile at scale and an agile organization. The right team is diverse and represents the division. The team needs to work in an agile way. In relation to having the right team it is also important to notice that SimCorp early on realized that they did not have the necessary knowledge of SAFe 4.0 internally. Thus they hired two companies to help them; one from Sweden (Crisp) and one from UK (Ivar Jacobson International). In total four consultants were paid for full time in about a year from early 2016 to early 2017. After the agile implementation and the first release of SimCorp Dimension by the agile organization SimCorp decided to hire agile coaches internally and built up a Lean Agile Office in SimCorp.

The first thing SimCorp would do differently if they were to redo the transformation to agile was to focus more on the technical challenge that lies in integrating new code into the existing code base – continuous integration. SimCorp Dimension is now an old product in that it has been in existence in approx. 20 years. That means that there is a lot of legacy code and it means that every time you integrate new code into the system you need to do a lot regression testing. When looking back (interview from March 2017) “we under-estimated this challenge”. All the other parts of SAFe were quite well documented but this part was not. “We should have had more focus on the individual developer’s toolbox”.

The other thing to do differently was to dedicate more attention to the change following new organizational roles. The change management process itself was planned using the ADKAR model (Hiatt 2006) an abbreviation for Awareness, Desire, Knowledge, Ability, and Reinforcement. The ADKAR-model talks a lot about the people side of change, about sponsorship, and about communication and there has been a lot of focus on that during 2016. But the impact of transitioning to new roles can never be underestimated. “We now have reinforcement left. All of 2017 is dedicated to reinforcement”

7 CONCLUSION

We will summarize the paper by looking back at the list of issues that started the process and discussing what has been solved and what not?

Agile and SAFe works as a way to improve systematically and continuously (issue A in the list above), manage distributed teams (issue H in the list), discover problems early (issue D in the list), and provides a lot of visibility (issue B in the list) about the software development.

Further agile with SAFe can handle the enormous complexity that SimCorp has as well as the huge amount of legacy code written into the Dimension product (issue F in the list).

Agile with SAFe seems to be better at reducing development risks (issue H).

It is planned to reduce the final quality and testing phase to one month out of six (thereby trying to deal with issues C and D) but that has not yet been realized partly because integrating new code into the existing code base was under-estimated (issue E).

Finally some people shows improved morale (issue I) but others are still confused about the new way of working. Thus the cultural change of the mindset of people takes a long time.

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